

G H RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT

(Approved by AICTE, New Delhi and Recognized by DTE, Maharashtra)

An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur

Accredited by NAAC with A+ Grade

Department of Electronics & Telecommunication Engineering

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AIM- IDENTIFYING REAL LIFE ENGINEERING PROBLEMS

Why These 5 Topics Were Selected Initially

The five topics were selected based on the following common criteria derived from the reviewed literature:

- Integration of IoT, Embedded Systems, and Machine Learning.
- Strong focus on real-world industrial or societal issues (Traffic, Infrastructure, Energy Efficiency).
- Alignment with emerging technologies (Smart Cities, Sensorless Monitoring, Spatial Mapping).
- Feasibility within academic project duration.
- Innovation and research publication potential.

1. Pothole and Plain Road Classification for Self-Driving Cars

Why it was considered:

- **Traffic Safety:** Road smoothness significantly impacts driving safety and comfort; potholes cause accidents and vehicle damage.
- **Advanced AI Integration:** Uses Deep Convolutional Neural Networks (ResNet-50) combined with a novel Adaptive Mutation Dipper Throated Optimization (AMDTO).
- **Automation:** Eliminates manual reporting by automating detection with high precision (99.79% accuracy).

Real-life applications:

- Autonomous/Self-driving vehicles.
- Intelligent Transportation Systems (ITS).
- Automated road maintenance and infrastructure monitoring.

It was selected because detecting road anomalies is a critical task for the safety and stability of next-generation autonomous vehicles.

2. Dynamic Real-Time Green Corridor for Emergency Vehicles

Why it was considered:

- **Societal Impact:** Delays in emergency vehicles (ambulances, fire brigades) due to traffic congestion lead to loss of life (20% of patients die due to traffic delays).

- **IoT Implementation:** Utilizes RFID tags, GSM modules, and a Central Traffic Management Server (CTMS) to create dynamic paths.
- **Optimization:** Proposes a "Dynamic Green Corridor" that clears traffic only where necessary (1-hop or 2-hop), minimizing disruption to normal traffic.

Real-life applications:

- Smart City Traffic Management.
- Emergency Medical Services (Ambulance clearance).
- VIP or High-Priority Vehicle routing.

It was selected due to its potential to save lives by reducing emergency vehicle travel time by up to 2.76 times.

3. Ultrasonic-Based Air Leak Detection in Factories

Why it was considered:

- **Energy Conservation:** Compressed air leaks in factories account for significant energy loss (10% of industrial electricity consumption).
- **Novel Methodology:** Uses spatial mapping (Ordinary Kriging) and ultrasonic sensors to visualize invisible leaks, superior to traditional manual inspection.
- **Industrial Relevance:** Addresses high costs and environmental consequences of energy wastage in large manufacturing plants.

Real-life applications:

- Large-scale manufacturing plants.
- Industrial pneumatic systems.
- Occupational health and safety monitoring (noise reduction).

It was chosen because energy efficiency in Industry 4.0 is a priority, and this method automates the detection of invisible energy losses.

4. IoT-Based Smart Street Light Fault Detection

Why it was considered:

- **Infrastructure Efficiency:** Addresses the inefficiency of manual inspections for broken streetlights, which compromise public safety.
- **Smart Features:** Integrates LDR sensors, GPS modules, and LoRa/GSM for real-time fault detection and precise location tracking.

- **Cost Reduction:** Reduces operational costs and energy wastage by ensuring lights are active only when needed and fixed immediately upon failure.

Real-life applications:

- Smart Cities lighting grids.
- Municipal infrastructure maintenance.
- Energy-saving automated lighting systems.

It was selected because it offers a scalable, IoT-based solution to a common urban problem, enhancing both safety and energy usage.

5. Sound-to-Vibration Transformation for Sensorless Motor Health Monitoring

Why it was considered:

- **Cost Innovation:** Proposes a "sensorless" approach by synthesizing vibration data from sound (using a phone) to detect motor faults, eliminating expensive accelerometers.
- **Advanced Deep Learning:** Utilizes 1D Operational U-Nets (Op-UNet) to transform audio signals into vibration signals for fault diagnosis.
- **Predictive Maintenance:** Allows for continuous health monitoring of industrial motors without complex wiring or sensor mounting.

Real-life applications:

- Industrial motor diagnostics.
- Predictive maintenance in aerospace and energy sectors.
- Remote machinery health monitoring.

It was chosen due to its highly innovative use of AI to reduce hardware costs in predictive maintenance.

Why 3 Topics Were Shortlisted from the 5

After evaluating all five topics, three were shortlisted based on:

- Practical feasibility within the lab environment.
- Availability of required hardware (Sensors, IoT modules).
- Direct societal or industrial impact.
- Complexity of implementation vs. time available.

The shortlisted three were:

1. **IoT-Based Smart Street Light Fault Detection**
2. **Ultrasonic-Based Air Leak Detection in Factories**
3. **Dynamic Real-Time Green Corridor for Emergency Vehicles**

Why "Pothole Classification" Was Not Shortlisted

- Requires large datasets and high computational power for Deep Learning models (ResNet-50).
- Difficult to simulate real-world road conditions and autonomous driving scenarios in an academic setting.

Why "Sensorless Motor Health Monitoring" Was Not Shortlisted

- Highly theoretical and relies on complex Deep Learning architectures (Operational U-Nets).
- Requires precise audio/vibration data synchronization and specialized motor test benches which may not be available.

Why Smart Street Light Fault Detection Was Finalized

Now comes the most important justification.

1. Global Problem with High Impact

- Many cities face poor street lighting maintenance due to manual monitoring.
- Faulty streetlights increase accident risk and crime rates.
- Energy wastage occurs when lights malfunction or stay ON unnecessarily.

This topic directly contributes to public safety, energy efficiency, and smart infrastructure development.

2. Strong Real-Life Applications

Smart street light monitoring systems are used in:

- Smart cities
- Municipal road lighting networks
- Highways and expressways
- Industrial campuses
- University and corporate campuses
- Residential societies

This makes the project highly practical and deployable in real environments.

3. High Research Depth

This topic combines:

- IoT sensor networks
- Embedded systems design
- Wireless communication (LoRa/GSM/Wi-Fi)
- Fault detection algorithms
- Real-time monitoring systems
- Cloud-based dashboards

It offers both hardware implementation and software system design contributions.

4. Innovation Possibilities

You can introduce:

- Adaptive brightness control using ambient light sensing
- Energy optimization algorithms
- Predictive maintenance using ML
- GPS-based fault localization
- Solar-powered smart lighting integration
- Edge-based real-time monitoring

This allows strong originality and technical innovation.

5. Feasibility Within Academic Duration

Compared to other topics:

- Hardware components are easily available
- No regulatory approvals required
- Easy to simulate using prototype models
- Moderate coding complexity
- Low implementation cost

It provides the right balance between technical depth and practical execution.

6. Publication and Career Value

This topic fits into research domains like:

- Smart Cities Infrastructure
- IoT-based Automation
- Sustainable Energy Systems
- Embedded Systems Engineering
- Intelligent Infrastructure Monitoring

It strengthens practical skills and increases chances of research publication or technical project recognition.

Final Structured Justification

Initially, five topics were considered based on their relevance to IoT and Machine Learning applications in modern smart systems. These topics addressed domains such as smart retail, healthcare, agriculture, infrastructure monitoring, and edge computing.

After comparative evaluation based on feasibility, implementation complexity, research depth, and societal impact, three topics were shortlisted: Predictive Maintenance, Smart Soil Monitoring, and Water Leakage Detection.

Among these, the IoT-Based Water Leakage Detection and Localization System was finalized due to its high societal importance, strong research potential, practical deployability, environmental impact, and integration of intelligent anomaly detection techniques. This topic not only addresses a critical global issue of water loss but also provides scope for innovative research contributions in IoT, machine learning, and smart infrastructure systems

Literature Review Summary Table

Author(s)	Publication Details (Year)	Findings of the Paper	Research Gap Identified
Ozer Can Devecioglu et al.	IEEE Access, Vol. 14 (2026)	Proposed a sound-to-vibration transformation method using 1D Operational U-Nets to synthesize vibration signals from audio. The synthesized signals achieved fault classification accuracy within 0.5% of real vibration data, eliminating the need for physical vibration sensors.	Acquiring reliable vibration data is costly and sensitive to sensor placement/mounting. Existing sound-based methods lack benchmark datasets and are often trained on small datasets, leading to biased results.
Amel Ali Alhussan et al.	IEEE Access, Vol. 10 (2022)	Developed a pothole classification system using Adaptive Mutation Dipper	Sensor-based methods are expensive and fail to detect potholes in

		Throated Optimization (AMDTO) and Random Forest. Achieved 99.795% accuracy, outperforming other optimizers and deep learning models (e.g., ResNet-50 at 93.8%).	the lane center. 3D reconstruction is computationally heavy, and standard image processing requires manual feature extraction. Deep learning requires large, balanced datasets.
Biru Rajak, Shrabani Mallick, Dharmender Singh Kushwaha	Procedia Computer Science, Vol. 171 (2020)	Proposed an IoT/RFID framework for dynamic green corridors. Found that a dedicated green corridor reduces emergency travel time by 2.76 times but disrupts normal traffic significantly. A 'one-hop' dynamic corridor was identified as the optimal balance.	Manual green corridors are inefficient. Sound-based detection has range limits and high false positives. ZigBee range (20m) is insufficient for timely clearance. Existing solutions lack cost-efficiency for real-time traffic management.
Leicai Xiao, Poorya Ghafoorpoor Yazdi, Sebastian Thiede	Procedia CIRP, Vol. 130 (2024)	Validated an ultrasonic leak detection method for factories using spatial mapping. Identified 'Ordinary Kriging with a spherical variogram' as the optimal mapping method (4.7% error). Proven effective in noisy environments where noise maps fail.	Manual leak detection is labor-intensive and sporadic. Acoustic methods fail in noisy factory settings. Automated ultrasonic methods exist for small pneumatic systems but are lacking for large-scale factory environments.
Sudha M et al.	IJLTEMAS, Vol. XIV (2025)	Designed a smart streetlight system using Raspberry Pi Pico, GPS, and sensors (LDR, voltage/current). Enables real-time fault detection (e.g.,	Traditional maintenance relies on manual inspection or complaints, leading to delays. Conventional timers/photocells lack real-time fault

		bulb failure, power outage) and precise location tracking via a web dashboard.	detection and location tracking capabilities, causing extended downtime and safety risks.
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Conclusion

After reviewing all four research papers, it is evident that IoT-based monitoring systems play a critical role in solving real-world infrastructure and safety challenges. The animal detection system contributes significantly to road safety; however, its impact is geographically limited. The IJIRT water monitoring paper provides a basic IoT framework but lacks advanced validation and performance analysis. The E3S 2023 paper introduces a sustainable flow comparison method, which is simple and effective but limited in intelligence and scalability. Among all, the IJNRD 2024 paper integrating IoT with Machine Learning demonstrates the most advanced and scalable approach. It incorporates multiple sensors, real-time alerts, GSM communication, and ML algorithms such as Random Forest for improved detection accuracy.

However, even this ML-based system has scope for improvement in terms of dataset expansion, sensor calibration, and large-scale deployment. Therefore, Water Leakage Detection remains the best research topic because it directly addresses global water scarcity, environmental sustainability, and infrastructure protection. It offers opportunities for integrating IoT, AI, cloud computing, and edge processing, making it technically rich, socially impactful, and highly relevant for future smart city applications.

References

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